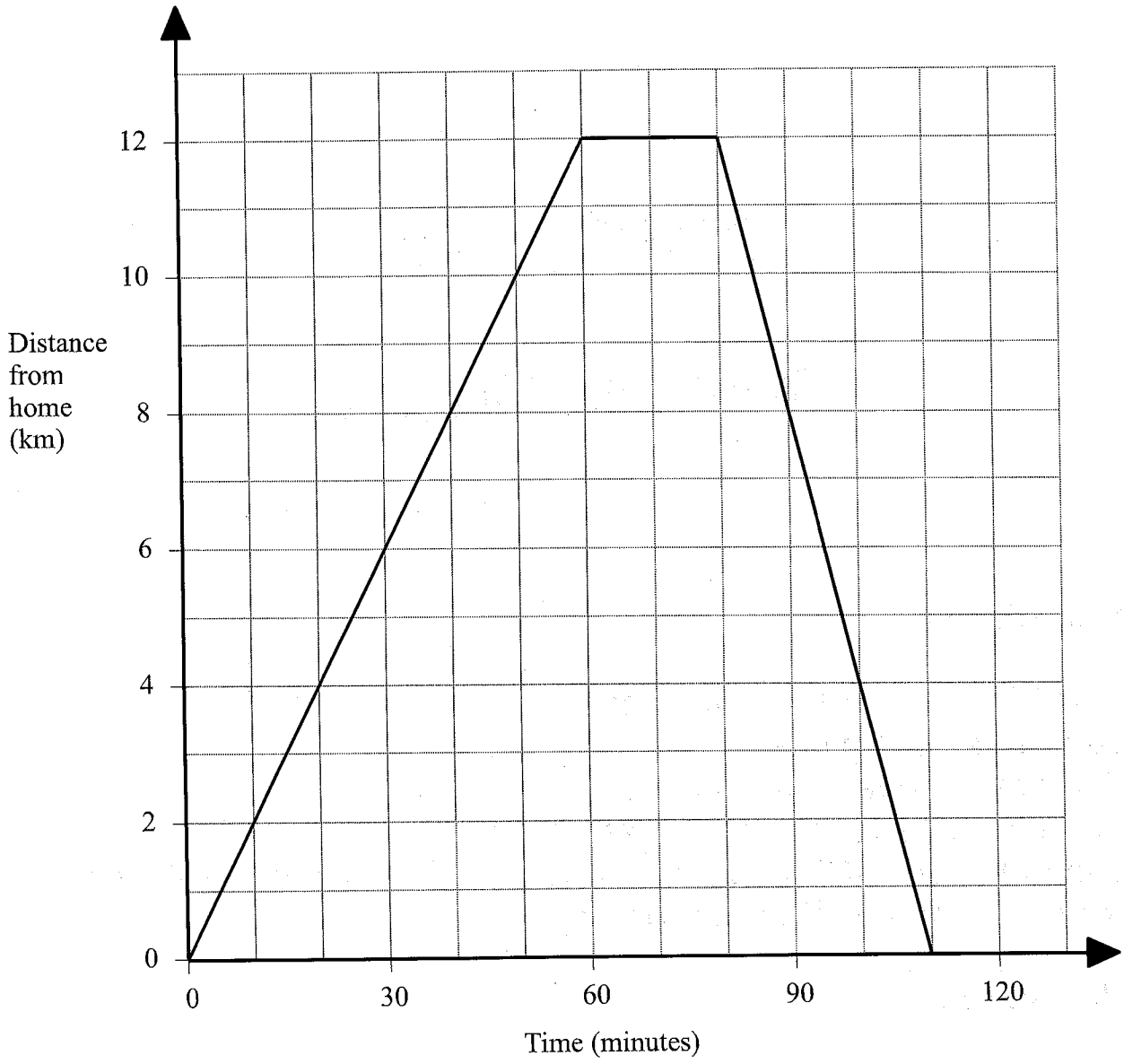
1 Rebecca went on a bike ride.
This graph shows Rebecca's distance from home on her bike ride.



(a) How far had Rebecca travelled after 30 minutes?

.....6.....km
(1)

(b) After 60 minutes, Rebecca stopped for a rest.
For how many minutes did she rest?

.....20.....minutes
(1)

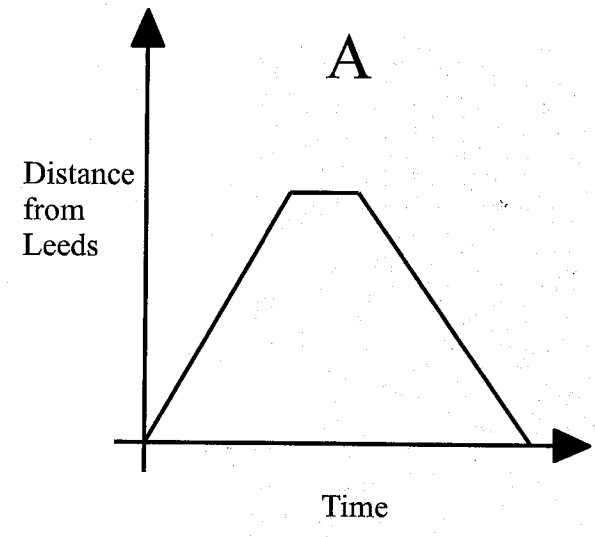
(c) How far did Rebecca travel in total?

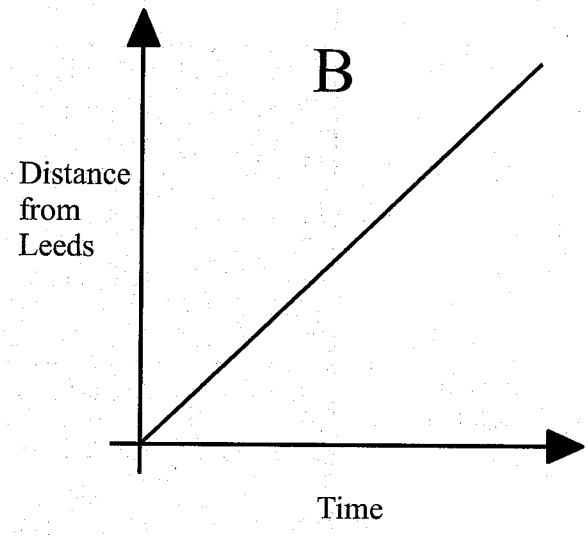
.....24.....km
(1)

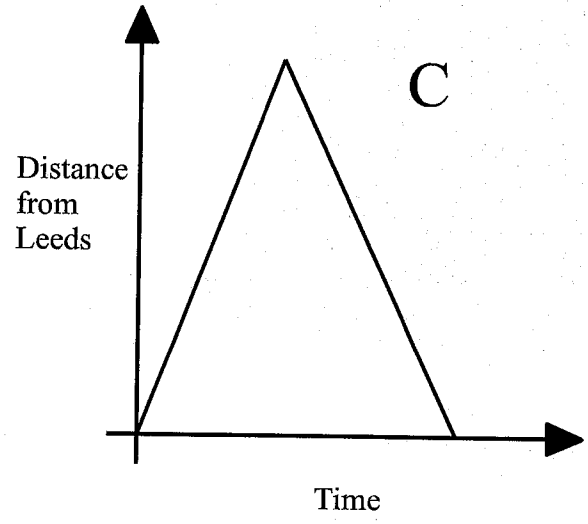
(Total for question 1 is 3 marks)

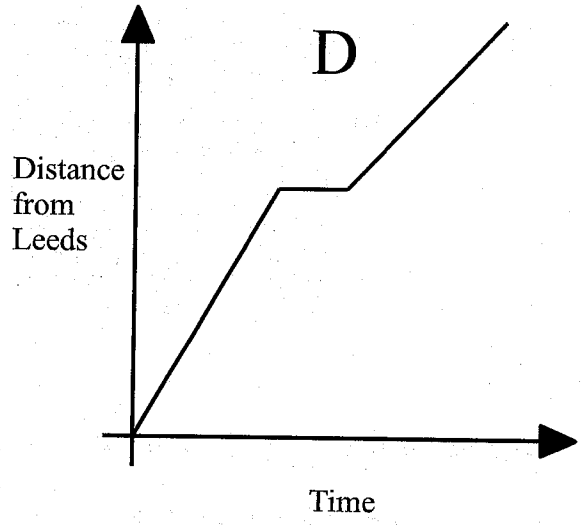
2 Jess drove from Leeds to Liverpool.
She stopped at a service station for 30 minutes on the way.

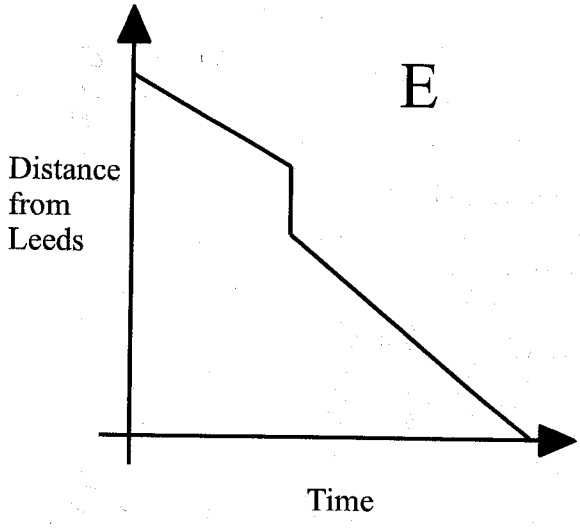
Which of these distance time graphs could represent Jess's journey?

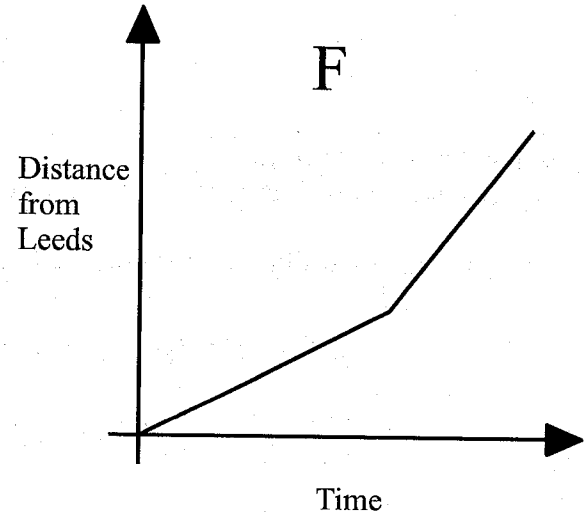










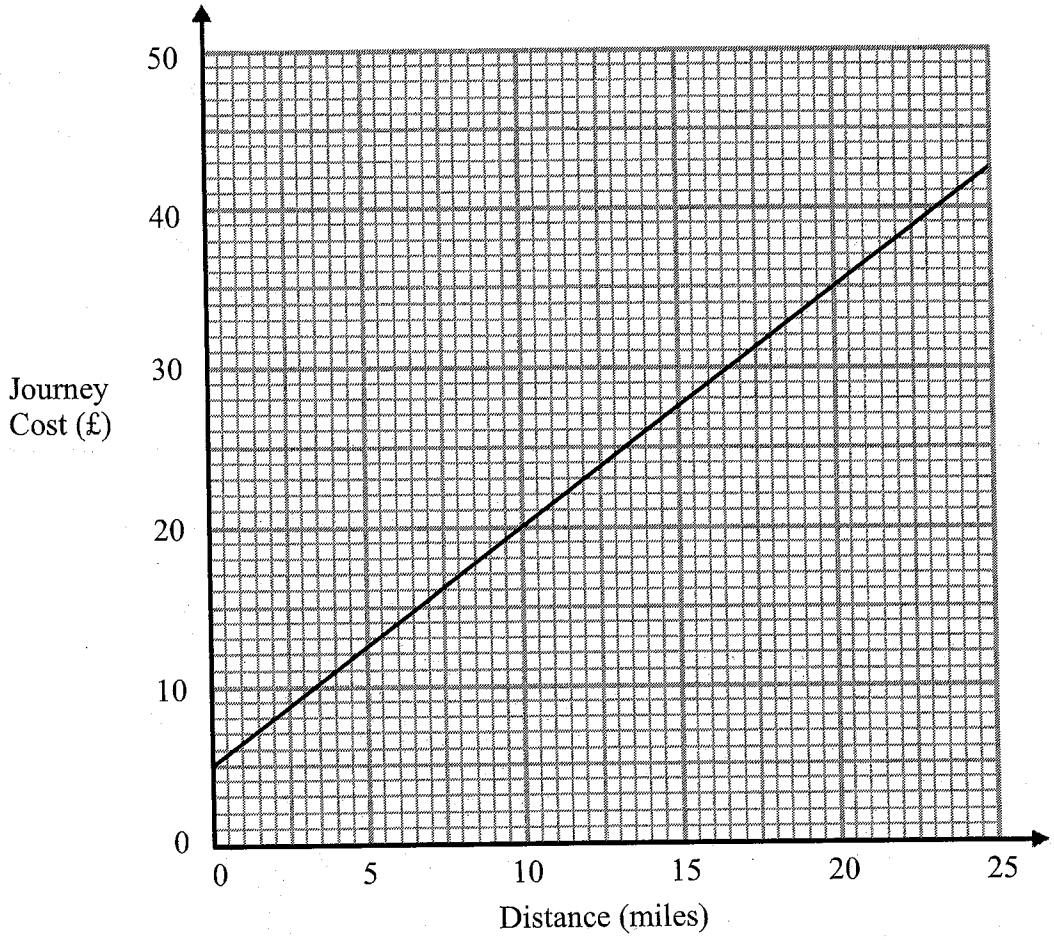


D

(Total for question 2 is 1 mark)

3 Bill is a taxi driver.

You can use this graph to find the cost of a taxi for different distances.



For each journey there is a fixed charge plus a charge for the distance.

(a) How much is the fixed charge?

£ 5

(1)

Bill makes two journeys.
The distance of one journey is 10 miles further than the other journey.

(b) Work out the difference between the two journey costs.

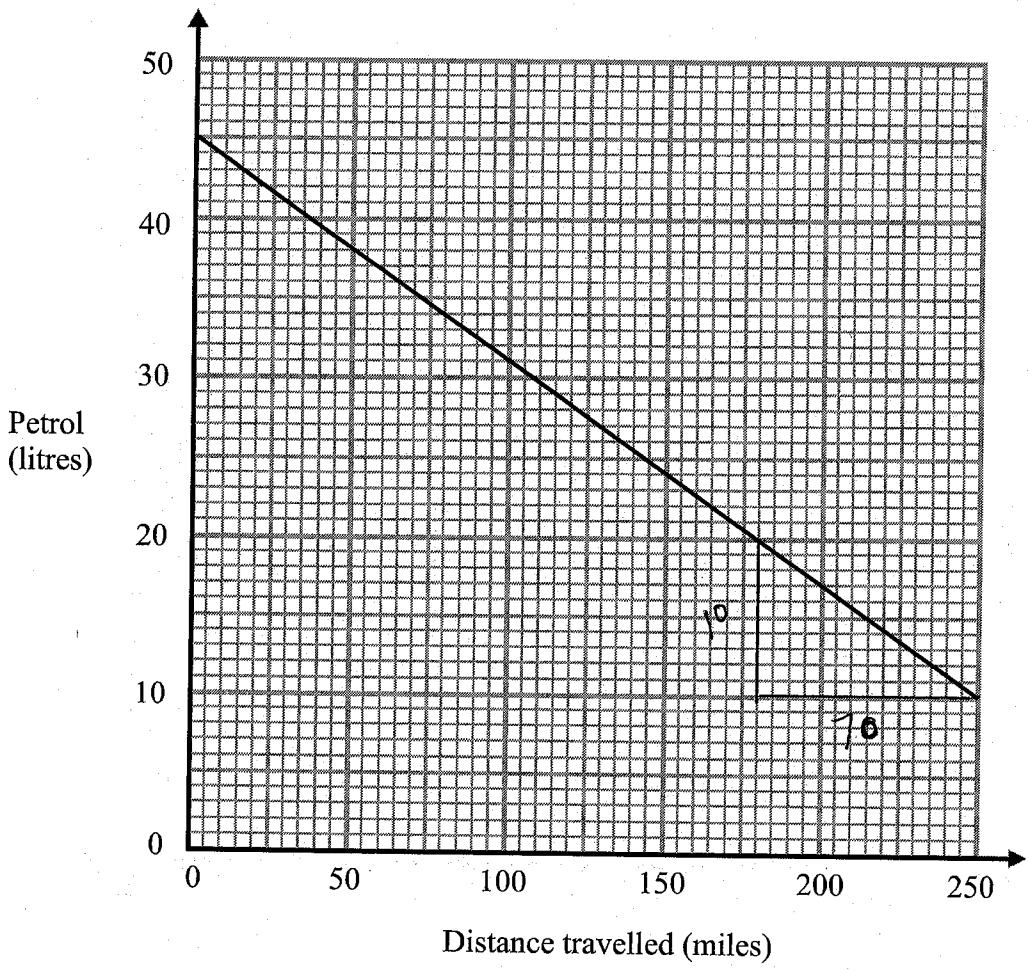
10 mile journey costs £20
20 mile journey costs £35

£ 15

(2)

(Total for question 3 is 3 marks)

4 John draws a graph to show the amount of petrol his car used on a 250 mile journey.



(a) How much petrol was in John's car at the start of his journey?

.....45.....litres
(1)

(b) How many more miles can John's car travel before it runs out of petrol?

.....70.....miles
[70 - 72] (1)

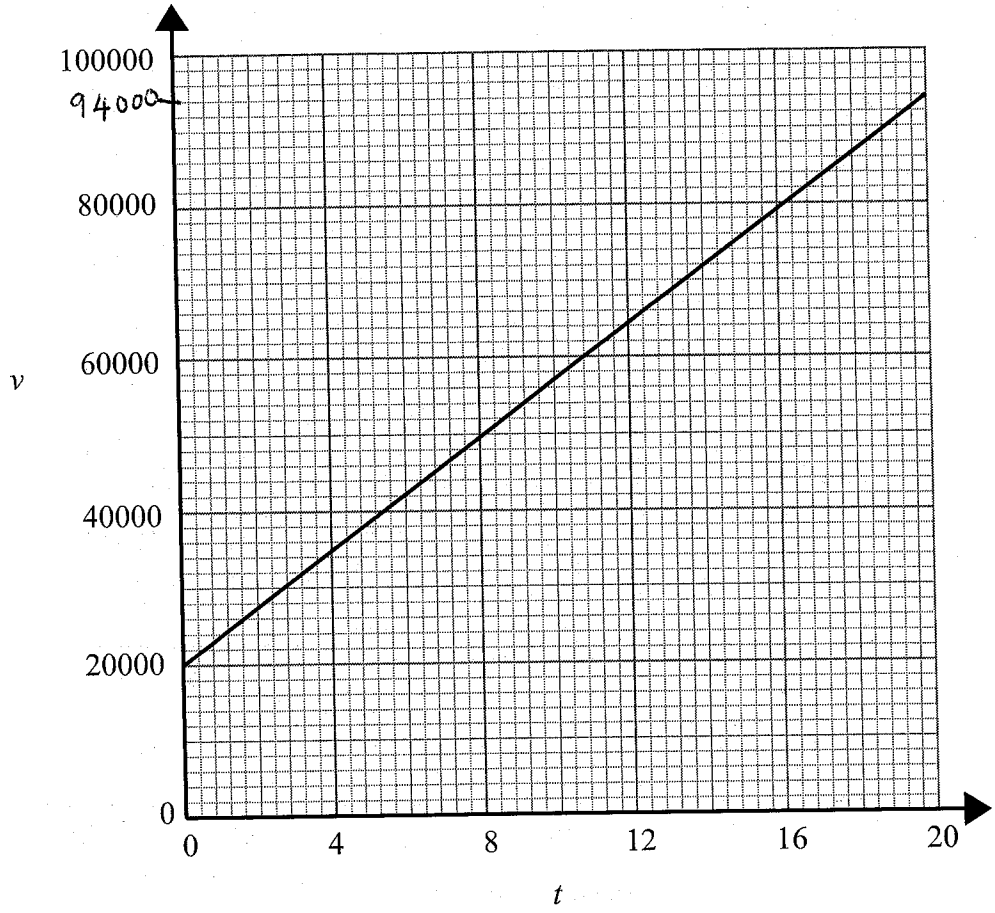
(c) What assumption have you made to calculate your answer to (b)?

.....petrol is used at the same rate.....

.....
(1)

(Total for question 4 is 3 marks)

5 Bernie is filling up a swimming pool.
The graph shows the volume v of water in the swimming pool at time t hours.



Work out the rate of that the water is flowing into the pool.

Give your answer in litres per hour.

$\frac{7400\cancel{0}}{2\cancel{0}} = 3700$

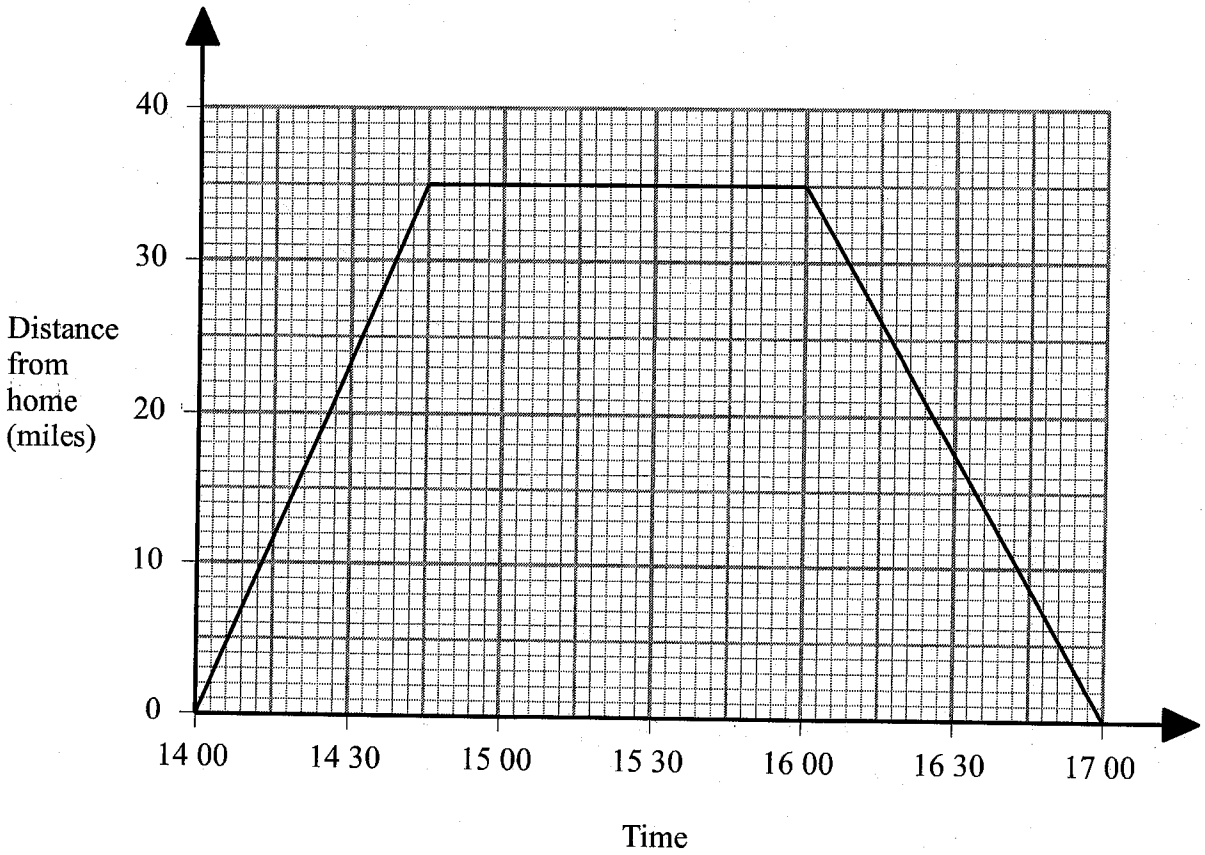
3700 l/hr

3700 - 3750 (2)

(Total for question 5 is 2 marks)

6 Emily drove to the beach. She stayed at the beach and then she drove back home.

Here is Emily's travel graph.



(a) For how many minutes did Emily stay at the beach?

1445 to 1600

.....75.....minutes
(1)

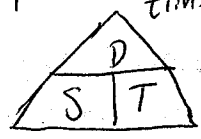
(b) What was Emily's average speed on her **journey to the beach**?

35 miles in 45 mins

0.75 hours

$\frac{35}{0.75} = 46.\dot{6}$ m/h

speed = $\frac{\text{distance}}{\text{time}}$



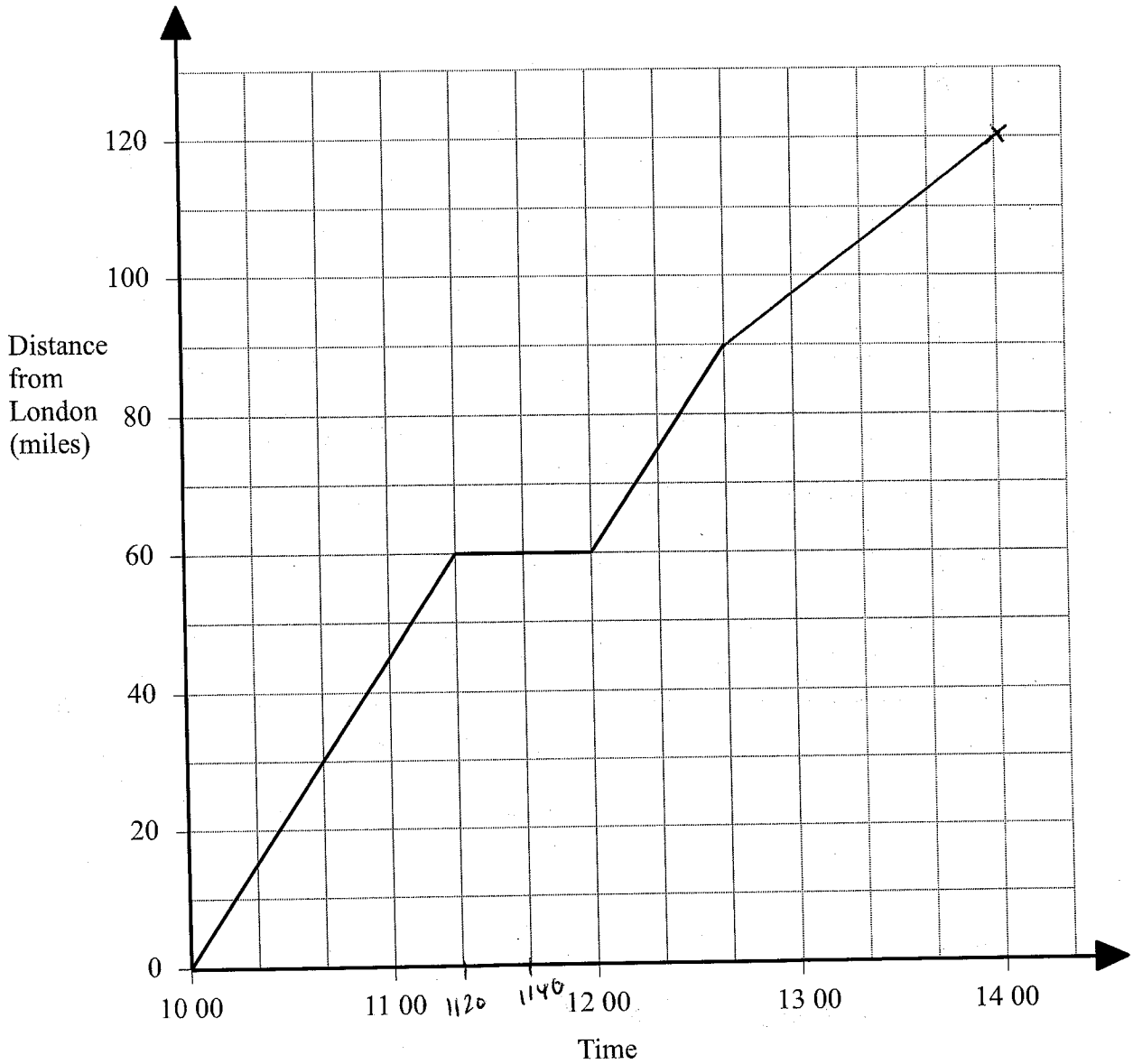
.....47.....miles/hour
(2)

(Total for question 6 is 3 marks)

[$\frac{140}{3}$ miles/hour]

7 Dawn drove from London to Birmingham. She made one stop at a service station.

Here is part of Dawn's travel graph.



(a) For how many minutes did Dawn stop at the service station?

.....40.....minutes
(1)

(b) What was Dawn's average speed between London and the service station?

60 miles in 80 mins

~~1.3 hours~~

30 miles in 40 mins

15 miles in 20 mins ~~60 ÷ 4/3~~

45 miles in 60 mins

.....45.....miles/hour
(2)

(c) Birmingham is 120 miles from London.
Dawn arrived in Birmingham at 14 00.

Complete the graph. (2)

(Total for question 7 is 5 marks)